



Transcript

Week 2: Mathematics and Statistics Fundamentals

Video 1: Mathematics and Statistics Fundamentals

Math and stats are everywhere, helping us solve problems, spot trends, and make smart decisions. But what do they really mean, and how do they work?

In this module, we will break down the basics and see how they apply to real world problems. From cracking numbers to uncovering patterns, you will see how math and stats are more than just calculations, they are the key to making smarter decisions.

Let's dive in!

Video 2: Vectors and Scalars

Speed tells you how fast, but direction tells you where! A car moving at 60 kilometre per hour north is a vector because it includes direction. On the other hand, speed alone, for example, 60 kilometres per hour is a scalar since it lacks direction. This difference is crucial in GPS navigation, which relies on direction, and in physics, where forces, acceleration, and momentum all depend on vectors. Understanding these concepts makes motion clearer in real life.

Scalars are quantities defined by magnitude alone, with no direction. They are represented using simple numerical values, such as 5 kg for mass or 30 degree Celsius for temperature. Other examples include speed and distance.

Vectors, however, have both magnitude and direction. Common examples include velocity, force, and displacement. They are typically represented with an arrow, where the length shows magnitude and the arrowhead indicates direction. Understanding this distinction is essential in fields like physics, engineering, and data analysis.

Scalars are represented as simple lines or numbers, while vectors are shown as arrows. The arrow's length indicates magnitude, and its direction shows where it points.

Vectors and scalars are essential in AI and machine learning. Vectors play a key role in algorithms that handle directions and forces, such as those used in robotics and computer vision. Scalars are equally important for measuring and setting parameters.

How do scalars and vectors shape the technology you use everyday? Think about it!

Video 3: Matrix and Operations on Matrix

How can a simple grid of numbers be the key to AI, graphics, and data science?

Let's explore what matrices are, how to perform basic operations, and their real-world applications.

A matrix is a rectangular grid of numbers arranged in rows and columns, making it an essential element in computer science, physics, and economics. From organizing massive datasets to solving complex equations, matrices are the foundation of countless real-world applications.



Let's explore basic matrix operations! You can add or subtract matrices of the same size, multiply them by a scalar, or perform matrix multiplication. Each operation follows specific rules essential for data manipulation. Here are visual examples: Matrix addition is done element-wise, scalar multiplication scales each value, and matrix multiplication combines rows and columns to transform data or represent relationships.

The **transpose** of a matrix flips it over its diagonal, converting rows into columns. The **inverse** of a matrix is a matrix that, when multiplied by the original, results in the **identity matrix**. Both operations play a crucial role in solving equations and data transformations.

This example illustrates the **transpose** of a matrix, where rows are converted into columns and vice versa.

The **inverse** of a matrix is also shown. Multiplying a matrix by its inverse results in the **identity matrix**. Both operations are fundamental in matrix algebra and play a key role in solving equations.

Matrices are essential in data science, serving as the foundation for data representation, transformations, and complex computations. They structure datasets by organizing observations and features, streamline calculations in linear regression, and form the backbone of neural networks by storing weight connections. In computer vision, matrices handle image transformations like scaling and rotation, while in NLP, they efficiently store word embeddings for language models.

Now, let's explore a real-world example: image transformation. An image is represented as a matrix of pixel values. By applying a transformation matrix, we can rotate, scale, or translate the image—key operations in image processing!

How can transformation matrices enhance image processing in your projects? Think about it!

Video 4: Determinants

Did you know that determinants play a crucial role in solving complex mathematical problems? From system equations to transformations in computer graphics, determinants help unlock powerful calculations. How can a single number reveal so much about a matrix?

In this session, we'll break down what determinants are, explore their properties, and learn how to compute them for different matrices. Let's dive in!

A determinant is a special scalar value derived from a square matrix. It provides key insights, such as whether the matrix is invertible and how it affects linear transformations. Determinants help in understanding matrix behaviour and are represented using notation as shown.

Determinants follow key rules that are essential in linear algebra. If a matrix has identical rows or columns, its determinant is zero. In a triangular matrix, the determinant is the product of its diagonal elements. Multiplying a row or column by a scalar multiplies the determinant by that scalar. Swapping two rows changes the determinant's sign. Finally, for matrix products, the determinant of the product equals the product of their determinants.



Calculating the determinant of a 2 by 2 matrix is simple. First, multiply the top-left and bottom-right elements. Then, multiply the top-right and bottom-left elements. Finally, subtract the second product from the first. For example, in a matrix with the values **3, 8, 4, and 6**, multiply **3 by 6** to get **18**, and **8 by 4** to get **32**. Subtract **32 from 18**, resulting in **-14**. The determinant helps determine if a matrix is invertible and provides insights into its properties.

Calculating the determinant of a **3 by 3 matrix** is more complex than a **2 by 2 matrix**. One common method is **Sarrus' Rule**, which applies only to **3 by 3 matrices**. This rule involves expanding along the first row and calculating the determinant using products of diagonals. By following this approach, we can systematically determine the determinant value, which is essential for various applications in linear algebra.

The determinant has a geometric meaning—it represents how a matrix scales space. In 2D, the determinant gives the area of the parallelogram formed by the matrix's column vectors. In 3D, it represents the volume of the parallelepiped. This concept is crucial in understanding transformations in linear algebra, with applications in fields like physics and computer graphics.

Determinants play a crucial role in data science. They help determine **matrix invertibility**, which is essential in solving linear systems using **Cramer's rule**. They are also used to compute **eigenvalues and eigenvectors**, which are key in **Principal Component Analysis or PCA** and machine learning algorithms. Additionally, determinants aid in **computer graphics**, helping with transformations and scaling. Understanding determinants is fundamental to advanced data science and machine learning applications.

Determinants are more than just numbers—they help us understand transformations, scaling, and invertibility in matrices. How can you apply these concepts in real-world problems? Keep exploring, keep learning!

Video 5: Role of Statistics in Data Science

How does statistics turn raw data into powerful insights?

Statistics is the foundation of data science, offering tools to **collect, analyse, interpret and present data** efficiently. It enables us to identify patterns and make well-informed decisions. **Descriptive statistics** summarise data, while **inferential statistics** use samples to predict trends in a larger population.

Statistics is essential in data science as it forms the foundation of data analysis and decision-making. It helps identify patterns, test hypotheses, predict outcomes, and clean data for accurate insights. Whether exploring data structures, validating assumptions, or preparing data for analysis, statistics plays a crucial role in every step.

Descriptive and inferential statistics serve different purposes in data analysis. Descriptive statistics summarise data, showcasing key measures like averages and variability. Inferential statistics help make predictions and draw conclusions about a larger population based on a sample. Understanding this distinction ensures the correct application of statistical methods.

Statistics is key in data science, powering techniques like regression, which identifies relationships between variables, classification, which sorts data, clustering, which groups similar points, and time series analysis, which tracks trends. These methods help in



predicting prices, detecting spam, and optimising business strategies.

A common real-world example is AB testing, where two versions of a webpage or app are compared to see which performs better. This helps businesses make data-driven decisions to enhance user experience and increase conversions.

Statistics is the backbone of data science, enabling us to analyse data, uncover patterns, and make data-driven decisions. By mastering descriptive and inferential statistics, you gain the essential tools to solve complex problems and drive meaningful insights.

But statistics is more than just numbers—it's about making informed choices and shaping the future with data. How will you apply statistical methods to uncover new opportunities and drive smarter decisions? Keep exploring, keep analysing, and keep improving!

Video 6: Types of Data

How does the type of data influence analysis and decision-making?

Understanding data types is essential in data science as it influences analysis, preprocessing, and model selection. The right approach ensures accurate insights and effective decision-making.

Let's start by understanding the main types of data. Data is generally classified into qualitative, also known as categorical data, and quantitative, which refers to numerical data.

Qualitative data describes categories or groups. It can be nominal, where categories have no natural order, such as colours or types of animals. It can also be ordinal, where categories follow a meaningful order, like education levels or satisfaction ratings.

Quantitative data represents numerical values and measurements. It can be discrete, involving countable numbers like the number of students in a class, or continuous, where values can take any number within a range, such as height, weight, or temperature.

Choosing the right data type is crucial for accurate analysis and visualisation.

Visualising data correctly is essential for effective analysis. Categorical data is best represented using bar charts or pie charts, making it easier to compare different categories. Numerical data, on the other hand, is often visualised using histograms, line charts, or scatter plots to show patterns, trends, and distributions. Choosing the right type of visualisation helps present data more clearly and accurately, allowing for better interpretation and decision-making.

Understanding data types is a crucial step in data science. It shapes how we preprocess data, choose analysis methods, and apply algorithms, directly impacting the accuracy of insights and predictions. Mastering this concept helps ensure data-driven decisions are both effective and reliable.

Now, how will you apply this knowledge to improve your data analysis? Keep learning and keep exploring!

Video 7: Descriptive Statistics

Have you ever stared at a spreadsheet full of numbers and thought—how do I make sense of all this?

Numbers alone don't tell the full story, whether you're analyzing sales trends, survey results, or business performance. That's where descriptive statistics comes in.

You'll learn key concepts like measures of central tendency and variability, making it easier to summarize and interpret data effectively.

Let's dive in!

Descriptive statistics summarise and describe the key features of a dataset. They provide simple summaries about the sample and the measures, helping us understand data distribution. These statistics are essential for initial data exploration, offering insights before deeper analysis.

Why are descriptive statistics important? They provide a quick overview of the dataset, helping to identify patterns, outliers, and anomalies. They are essential for data cleaning and preparation, guiding decision-making in data processing. By simplifying large datasets into understandable summaries, descriptive statistics support better data-driven decisions.

Measures of central tendency help identify the centre of a dataset. The mean, or average, is calculated by summing all data points and dividing by the number of values. The median is the middle value when data points are arranged in ascending order, while the mode is the most frequently occurring value in a dataset.

Let's calculate the mean of 5, 10, 15, 20, and 25. First, we add all the numbers, giving a total of 75. Since there are 5 numbers, we divide 75 by 5, resulting in a mean of 15.

Now, let's determine the median of 1, 3, 5, 7, and 9. When arranged in order, 5 is the middle value, making it the median.

Lastly, let's find the mode of 2, 3, 4, 4, 5, and 6. Since 4 appears most frequently, it is the mode.

These measures provide valuable insights into data distribution.

When should we use each measure of central tendency?

The **mean** is best when the dataset has no extreme outliers, as it provides a balanced average.

The **median** is useful for skewed distributions or datasets with outliers. For example, when analysing **house prices** in a neighbourhood with a few very expensive homes, the median gives a more accurate central value.

The **mode** is ideal for **categorical data** or when identifying the most common item. For example, in retail, the mode helps determine the **most frequently sold shoe size** in a store.

Each measure is valuable, depending on the dataset's nature and the type of analysis required.

Measures of variability play a crucial role in data analysis by indicating the consistency of data points. A **low standard deviation** means that the data points are close to the mean, suggesting high consistency. In contrast, a **high range or variance** indicates greater spread, which may suggest data variability or the presence of outliers. Understanding these measures helps assess the reliability and distribution of data, making them essential for accurate analysis and decision-making.

Descriptive statistics are widely used across industries to analyse data patterns and variability. In **business**, they help track average sales and fluctuations to predict future trends. In **healthcare**, they are used to monitor patient vitals, such as average heart rate and its variability over time, ensuring better health assessments. In **education**, they assist in evaluating average test scores and their spread, helping measure student performance.

and identify learning gaps. These applications make descriptive statistics essential for data-driven decision-making.

To summarise, descriptive statistics offer a clear snapshot of a dataset by highlighting central tendencies and variability. Selecting the right measures helps in understanding data distribution, detecting patterns, and preparing for deeper analysis. These tools are essential for identifying trends, spotting anomalies, and making informed decisions.

Now that you understand descriptive statistics, how will you apply them in your work or studies? Keep exploring data to uncover insights and improve decision-making.

Video 8: Introduction to Probability

How do we measure uncertainty and make informed decisions?

Probability provides a systematic way to quantify uncertainty, helping us predict outcomes and assess risks. By understanding its fundamental concepts, definitions, and rules, we can make more accurate predictions and informed choices in various fields.

Probability measures the likelihood of an event occurring. It quantifies uncertainty by expressing how likely an outcome is using values between 0 and 1. A probability of **0** means the event will not happen, while a probability of **1** means it is certain to occur. Values between 0 and 1 indicate different levels of likelihood, helping us assess and predict outcomes in various situations.

Let's explore some key terms in probability. An experiment is a process that leads to one of several possible outcomes, such as rolling a die. An outcome is a specific result of an experiment, like rolling a 4. An event is a collection of one or more outcomes, such as rolling an even number. The sample space, S , represents all possible outcomes of an experiment, for example, 1, 2, 3, 4, 5, 6 when rolling a die. Understanding these terms is essential for analysing probability.

Probability can be classified into three types. Theoretical probability is based on the assumption that all outcomes are equally likely. For example, the probability of getting heads in a fair coin toss is 0.5. Empirical probability is determined through observation and experiment. If a coin is flipped 100 times and lands on heads 48 times, the empirical probability is 0.48. Subjective probability is based on personal judgment or experience, such as estimating the likelihood of rain tomorrow based on past weather patterns. Each type of probability provides a different way to assess the likelihood of events.

Let's go over some basic probability rules. First, the probability of any event is always between 0 and 1, meaning it cannot be negative or exceed 1. Second, the sum of the probabilities of all possible outcomes in a sample space is always 1. This ensures that all possible scenarios are accounted for. Lastly, the **complement rule** states that the probability of an event **not occurring** is **1 minus the probability of it occurring**. For example, if the probability of rain is 0.3, then the probability of no rain is **0.7**. These fundamental rules help in calculating and interpreting probability effectively.

Probability helps us determine the likelihood of events occurring, and two fundamental rules guide these calculations: the addition rule and the multiplication rule.

The addition rule applies to mutually exclusive events—those that cannot happen at the same time. To find the probability of either event occurring, their individual probabilities are added. For example, when rolling a die, the probability of getting a 2 or a 5 is found by adding the probability of each outcome, resulting in $1/3$.

The multiplication rule is used for independent events, where one event does not affect the other. The probability of both events occurring is calculated by multiplying their individual probabilities. For instance, when flipping a coin twice, the probability of getting two heads in a row is determined by multiplying the probability of heads on each flip, resulting in 0.25.

These rules are essential for predicting event likelihoods and making data-driven decisions.

Conditional probability tells us how likely an event is, given that another event has already happened. This concept is important when one event influences the probability of another. The formula for conditional probability is:

Probability of A given B equals Probability of A and B divided by Probability of B.

For example, in a deck of fifty-two cards, the probability of drawing an **Ace, given that the card drawn is a Spade**, is:

Probability of Ace given Spade equals Probability of Ace and Spade divided by Probability of Spade.

This equals one by fifty-two divided by thirteen by fifty-two, which simplifies to one by thirteen.

This concept is essential in probability analysis, especially when dealing with dependent events.

Probability has many real-world applications across different industries.

In business, it plays a key role in risk assessment and decision-making, such as setting insurance premiums or analysing stock market trends.

In healthcare, probability helps in diagnosing diseases by analysing test results. For example, it can be used to determine the probability of a positive test result for a specific condition.

In gaming, probability is essential for understanding odds in games of chance, such as poker or blackjack, helping players make more informed decisions.

Probability is a powerful tool that helps industries manage risks, improve predictions, and make strategic decisions.

To summarise, probability allows us to quantify uncertainty and make better decisions. Understanding its rules and types is crucial not just in data science, but also in everyday decision-making. Probability plays a key role in business, healthcare, and gaming, shaping strategies and predictions across industries.

Now, think about how can probability improve your decision-making?

Video 9: Probability Distributions

Ever wondered how we can predict stock prices, weather patterns, or even dice rolls? It all comes down to probability distributions.

Probability distributions help us understand, model, and predict data patterns. By analysing different types of distributions, we can determine how data points are likely to behave, making probability a powerful tool in decision-making. Understanding these distributions is key to making data-driven predictions and managing uncertainty effectively.

Let's start with the basics. A **probability distribution** describes how the values of a **random variable** are spread out. It represents the possible **outcomes of an experiment** and the **probability assigned to each outcome**. These distributions are essential for **understanding data patterns** and **modelling real-world phenomena**, helping us make predictions based on uncertainty.

Probability distributions are classified into two main types: **discrete and continuous**.

Discrete probability distributions apply to variables with distinct, separate values. For example, the number of heads in a series of coin tosses follows a discrete distribution since the outcome is limited to whole numbers. **Common discrete distributions include the binomial and Poisson distributions.**

Continuous probability distributions apply to variables that can take any value within a range. For instance, the heights of people in a population follow a continuous distribution since height can be measured to infinite precision. **Common continuous distributions include the normal and exponential distributions.**

Understanding these distributions helps in modelling real-world data and making informed decisions.

Probability distributions are classified into two main types: discrete and continuous. Discrete probability distributions apply to variables that take distinct, separate values. The binomial distribution describes the number of successes in a fixed number of independent trials, such as counting heads in a series of coin tosses. The Poisson distribution models the number of times an event occurs in a fixed interval, like the number of emails received in an hour. Both distributions are essential for analysing and predicting the frequency of events and occurrences in various real-world contexts. On the other hand, continuous probability distributions apply to variables that can take any value within a range. The normal distribution, also known as the Gaussian distribution, is symmetric around the mean and describes many natural phenomena, such as heights or IQ scores. The exponential distribution models the time between events in a Poisson process, such as the time between bus arrivals. Understanding these distributions is essential for modelling real-world data and making accurate predictions.

Distributions have key characteristics that help in understanding data. The **mean** represents the central value, while **variance** measures the spread of data points. **Skewness** indicates the asymmetry of a distribution, showing whether data is skewed left or right. **Kurtosis** describes the tail heaviness, helping identify whether a distribution has extreme values. These characteristics are essential for analysing patterns and making data-driven decisions.

Probability distributions are widely used across various industries. In finance, they help model stock prices and assess risk using normal and log-normal distributions. In healthcare, they assist in analysing the distribution of patient vitals, such as blood pressure and cholesterol levels. In manufacturing, they ensure quality control by modeling defects and failures using binomial and Poisson distributions.

Choosing the right **probability distribution** depends on the nature of your data and the assumptions you can make.

Use **discrete distributions** for **countable outcomes** and **continuous distributions** for **measurable quantities**.

Understanding these distinctions is crucial for accurate **data analysis and interpretation**.

Different distributions serve specific purposes based on the **data type** and **scenario**. Knowing which one to use ensures effective **modelling and decision-making**.

Thank you for exploring the key concepts of probability distributions.



Understanding the right distribution for your data is essential for accurate analysis and decision-making.

Now, it's your turn! How will you apply these insights to your data analysis?

Keep learning, keep analysing, and make data-driven decisions with confidence!

Video 10: Vectors- Demo

Hello and welcome everyone. Let us go through the concept of vectors. In the today's presentation, you will observe that we are going to demonstrate how vectors look.

You also have a Python code to generate vectors. But since we have not covered yet Python, I would want you to not get overwhelmed with the Python code. In fact, just use the Python code for intuition and the docstring which is mentioned here explains about what a particular function is doing.

So, if you can follow me closely, we would be able to evidence the way in which the vectors are plotted with the magnitude and the directions and the different types of vectors in this lesson. So, let us start with the simple definition of a vector. We know that a vector contains a magnitude and a direction and that is exactly what I mentioned.

We are trying to plot a two-dimension vector with a magnitude and direction. Now, because it is a two-dimension vector, you see we have an x and a y component of that and I am simply going to plot the x and y component. When I go ahead with plotting the component, I say that the x component is 3 and the y component is 4. You can also try to replace that with any other number and I try to plot that vector.

If you see this is a 3 and a 4, 3, 4 vector, it is a position vector and you see originates from the origin and it goes all the way to 3, 4. And by the definition of magnitude, the way in which we calculate the magnitude, you will see that we come up with this idea of magnitude being 5. So, this is a classic example where you see we can plot a vector and if I say 3, 4, this is what I need to say, this is what it looks like. Now, this is a two-dimension vector. What if I just, what if I am interested in one dimension vector, we call that as a row or a column vector.

If I try to plot a row vector, it will just be a single row vector, it will be a one-dimensional array and it will still have magnitude direction. So, almost very similar in the idea as we had done above, but earlier we had two components, now we will extract the number of components from the row vector. If I pass the example 3, 4, you will see that since it is a row vector, so I am interested in the y component of that.

So, this is basically the y component, the magnitude still is 4, but I am interested in the y component, so I capture the y component. For a similar vector, if I plot the column vector, you will see that I will have plotted it in the column wise manner. So, the same vector in the column wise manner, the same vector in the row wise manner.

So, that is how if someone speaks about a row vector or a column vector, you should be able to imagine that as a vector which is horizontal if I speak about a row vector and if you speak about a column vector, it will be vertical like this. So, that is essentially what the two different types of vectors are.

There is a third and a very common type of a vector, we call that a unit vector. Now, a unit vector is special because its magnitude is 1 and it has some direction. It is still a two-dimensional vector, it is just that the magnitude of that vector will be 1. Now, if you see here, we are extracting the magnitude and then we are normalising it to make it a unit vector.

I further plot that and if you see a classic example of a unit vector, it still originates from the origin and if you see the magnitude of this vector, it is exactly as 1. This is a classic example of unit vector and in practise, we use unit vector to specify the direction of a resulting vector. So, we have seen a vector, a row vector, a column vector and a unit vector. Further, there is something called as a position vector.

Position vector is no different from the first picture which you saw, where we try to, if I am plotting a position vector for 3, 4, it will be nothing but it originates from the origin and it will go all the way to 3 on the x axis and 4 on the y axis. So, that is the reason when I was showing you the first picture, I alternatively use the name position vector. This is very important because this helps us in performing a lot of vector equations later.

Now, we have seen that all of those had origin as the point where they initiated. What if I want to change the reference point? Now, if I want to change the reference point, I will just provide the reference point example. Let us say I still want to plot the same position vector 3, 4 as I have been doing till now, but I want to change the reference point to a different reference point 2, 3. Remember, up until now, we were using origin as a reference point which was 0, 0. Now, when I see this, you see that it does not originate from the origin anymore.

It is originating from 2 and 3 and still goes all the way to 3, 4, but because it is originating from a different reference point, you see the magnitude changes. We had the magnitude as 5, but now we have magnitude recalculated as 1.41. So, that is how you will observe the change in magnitude if you change the reference point. There is another concept called as displacement vector, very interesting concept.

What are we going to do here? We are going to plot the displacement vector and try to study its magnitude and direction. There will be a starting point and there will be an ending point. And then we will define the displacement vector as the end point minus starting point.

How much did the vector get displaced by? Now, if we skip this and we just jump to the example usage, this is the starting point and this is the ending point. And if I plot the displacement vector, this is how it would look like. You see it still has the magnitude 5, but this is a classic example of a displacement vector where I started at point 2, 3 and I ended up with being at point 5, 7. And you see here, this is 5, 7. So, that is how we tend to plot a displacement vector.

Vector operation follows certain properties. For example, a simple property is a commutative property. So, I am going to take two vectors and sum them to demonstrate that the vector follows commutative property.

So, you see here, I am going to take 2 itself. So, I am going to take u as 2, 3 and v as 1, 1. And I am going to plot $u + v$ and $v + u$. So, if you see here, this is my u , this is my v and this is my $u + v$. And if you plot the same way by doing $v + u$, you will still see the same vector.

So, you see here, this was my red, now this is the red, here this was my blue, now this is where it is blue. So, we just interchange the position, but we are still able to generate the same resulted vector that is $u + v$, which is green. And you can see that it follows 3, 4 here follows 3, 4. So, that is a visualisation of the vector addition following the commutative property.

Further, if you see the associative property, where we are going to demonstrate that how we try to prove $u + v$, $v + w$ and $u + v + w$ and $u + v + w$. If you see here, just compare this picture and this picture. So, you see that the resulted picture, this picture and this picture, the resulted vector that is this green vector 4, 9 and here also it is 4, 9. So, the resulted vector which is, which follows $u + v + w$ and here it is $u + v + w$. So, we have calculated $u + v$ and $v + w$ separately, so that we can add those vectors. So, this is a classic example, a proof that vectors they follow the associative property.

If I go ahead and I try to detect the 0 property, what is 0 property? So, let us try to see, I have a vector `np.array(2, 3)`. I just try to add a 0 to it, if I add the 0, you will see that it will be the same vector, it will result in the same vector. So, there is an overlap, that is why we do not tend to see the 0 vector and the vector v , but it is just completely overlapping with $v + 0$. So, v and $v + 0$, they are the same. So, $v + 0$ is equals to v , that is what 0 property says.

Further, there is another property called as additive inverse. So, if I am going to plot a vector and it is additive inverse, it will show us a resultant vector. Let us try to see the resultant vector.

So, I have a vector 2, 3. I am going to add the additive inverse of that and I will observe the resultant vector to be 0. Let us try to do that. I have a vector v which is blue in colour and I have its additive inverse which is red in colour. And if I sum them up, you see the green which is $v + \text{minus } v$, it is actually nowhere to be found, the reason because its magnitude is 0. And hence we call that as an additive inverse, exact opposite of vector v . If you add that, it will completely get vanished.

The same way we had the additive associative property, the same manner we have the multiplicative associative property $A * B * v$ and $A * B * v$. If we try to do that calculation, we try to prove that they are going the same. So, now I will jump to the, you see we have $A * B * v$ which is this vector and $A * B * v$ which is this vector. If you compare them, they will exactly be the same.

In a way, they are copy of each other which means that these operations are equal and they follow the scalar associative property. We had seen the additive associative property above and here we are now observing the scalar associative property. Similarly, we will also be able to observe the distributive property.

That is, if I have a scalar and I multiply that with $v + B * v$, it could also be done in this manner. So, we try to plot that, we are trying to prove this theory. So, I am going to just show you the chart where we have the resultant vector on the LHS and of the RHS.

If you see here, this is what we have the resultant vector of the LHS and this is what we have of the RHS and if you see there is that. Another property which vector follows is the scalar multiplicative identity property. And it is very simple, say if you multiply the vector by an identity vector, it would remain as it is.

If you remember, this was 0 property, when you added a 0 vector, it was as it is. But in multiplication, you are going to multiply that by a unit vector or a magnitude with unit vector. So, this is my original vector and I multiply that by 1, I get the resultant vector which is exactly the same.

Now, what if I want to plot the original vector, but it is a scalar multiples to demonstrate that how the magnitude changes. So, observe here, I have this vector 2, 3 and I am going to plot the twice of that vector. This is my original vector, the direction is the vector is following this direction, but the magnitude is 3.6. But when I multiply that by 2, the direction still remains the same that the magnitude changes to 7.2, which is a multiple of



3.6, twice of 3.6 will be 7.2. You want to try that with 3, 4, 10, 20, any number, you would observe that the direction would remain the same, but the magnitude would change.

Similarly, if you perform vector addition, you will observe that if you take two vectors, vector v_1 and vector v_2 , and if you try to add that the resultant vector would be this green vector v_1 plus v_2 . It follows the formula, if you remember the vector addition, it will basically try to follow the rectangular formula. So, this would be a diagonal of the rectangle, if you form a rectangle, a complete rectangle, this would be the diagonal of that rectangle.

Similarly, if you want to do vector subtraction, vector subtraction is nothing but addition of a negative vector. So, for example, if I have v_1 and v_2 , if I were to add v_1 and v_2 , you will see that this would be the resultant vector. Earlier, if you remember, this was the resultant vector, the diagonal, but now this time, this would be the vector v_2 .

Another way of doing it is to plot the vector subtraction with reversing the vector. So, for example, see this was my vector v_1 , this was my vector v_2 . So, what am I doing is, I am reversing the direction, see the direction, it is in this direction, I am reversing the direction.

After reversing, I join the endpoints and that is my v_1 minus v_2 . So, we saw vector addition and subtraction. So, this diagram basically talks about addition and subtraction in one picture.

We have the original vector, if you add that, you will get the resultant vector like this, subtract that, you get the resultant vector like this. Let us move to another crucial concept, the dot product. When you are trying to take two vectors and trying to multiply them, you can use that the dot product concept behind that.

So, I take a vector 1, I take a vector 2, I simply calculate the dot product, it results in a scalar value which is 8. Another way of multiplying two vectors is cross product, where if you try to multiply the two vectors, it will result in a vector. You do not need to bother about the Python code here. Once we cover Python, it will make more sense.

Another concept is the norm of the vector. So, for example, I have this vector, I want to calculate the normal vector, it requires me to tell the value of p . I want to calculate the 3 norm of the vector, I can mention the p example as 3 and this is how it is calculated to be 3, 3. For a vector 2, 3, if you want to calculate the p norm, this is how the calculation happens, cube of 2 plus cube of 3 and cube root of the sum of the multiplicants, the power of the each coordinate. You can take another example and try to apply a 2 norm and that would be 5. I want all of you to quickly recall, where did we see the vector 3, 4 and how is 5 associated with that.

If you quickly recall that, we should have seen this somewhere about. Similarly, you can take norm of different vectors just by passing the value of p over p . You might have heard about any Euclidean norm of a vector, which is nothing but you tend to use the value of p as 2 over there and this is how you calculate the Euclidean norm for the same vector 3, 4. So, we had the norm as 5, this Euclidean norm which is squared and 2 norm will be 5 square which is 25. Similarly, you can also calculate the L1 norm of a vector.

So, if you have this vector, calculate the L1 norm, it should be 8. You slightly add a noise epsilon $1e-6$, which is 10 to the power of minus 6, the L1 norm slightly changes. You take another example and try to calculate the p norm mentioning the value of p , get 16 and for the same vector, value of 2 will be 9.16. We tend to use these norms in vector operations later. So, you can get yourself accustomed to these visualisations and when we use them later, you would be able to relate that.

That is all for vectors from my side. Thank you everyone.

Video 11: Descriptive Statistics- Demo

Hello and welcome everyone. In the today's session, we are going to go through a very crucial topic, especially in the context of data science, artificial intelligence, and machine learning, and we call that as descriptive statistics.

You know that whenever we talk about data collection of information, we always use statistics as a tool to talk about collecting the data, analysing it, interpreting it and presenting it to the right setup stakeholders.

So, all the set of methods, tools, techniques which can help me to understand the data, to extract insights from the data, we can term that as statistics.

Now the moment you decide to explore the data; you do that by understanding the description of the data and that's where the name descriptive statistics comes from. What are those measures which can help me understand the data very well?

In summary, so when we talk about statistics in data science, there are two types of statistics which are very commonly used.

Descriptive statistics where all the methods which helps me to understand what the data is talking about is descriptive statistics. Further, inferential statistics is about once you have understood what inferences can you make from the data.

A classic example if I give you, let's say there is an election process in a certain country, to understand how many people would participate, to understand the number of males, the number of females who would participate, age wise study, profession wise study, inclination towards certain party wise study is all which comes under descriptive statistics.

Once I have studied this then I will try to make inferences that which party would win the election. So that is basically where I am claiming where I am coming up with a hypothesis for a certain scenario that is inferential statistics. So, you know that this is all done on data and data is nothing but any information which can be translated into some useful insights which can help you to make a certain decision.

And when we talk about data, you know that data is broadly classified into numerical and categorical.

From the name itself it suggests that numerical is made of numbers, age, weight, number of children, temperature, shoe size and categorical is where it is made of categories, usually nonnumeric or categorical.

For example, colour of an eye, the blood type, gender, ethnicity, nationality. Within numerical further you have division. Is it continuous? Is it discrete? Age, weight, blood pressure, temperature these are continuous. Shoe size, number of children, these are all numerical, discrete and within categorical. Again, you have two types.

Where there is a certain order, there is a certain hierarchy. We call that as ordinal. For example, the severity, pain, satisfaction, mood, size of a glass. Nominal where you don't have any hierarchy, where you cannot arrange the categories in any order, colour of an eye, type of blood, breed of a dog, you don't have any hierarchy, any order over here.

So, we saw that statistics is about Oregon, Descriptive statistics is about trying to use methods which can describe, summarize my data in a way that I can confidently say that I know what the data is about.

Now in statistics there are two terms, population and sample. Population is nothing but that collection of individuals or objects or events who is under study. For example, when I gave you the election example in that the entire country is the population. Let's say we are studying the distribution of male students versus female students. Then my population is all the students. So, whatever is under study, under analysis is my population. But obviously you know that you cannot when you are try to in the election process, when you try to study which party will win the election, you don't go and meet each and every person for you to make an inference.

You try to sample a subset of the population and that's what we call as sample. Now, whenever we want to make any inferences about the population, we study the sample, and we assume that whatever we are going to make an assumption of a sample would also apply to the population. For example, if I want to calculate the mean, which is an average, I will calculate the average of the sample population and I would assume that it is also applicable to the larger audience, the population, entire population.

So, the first measure in descriptive statistics is mean, which is simply the average of 2 numbers. And you know that you use mean to understand the central tendency, which is that value where the data revolves around. For example, if I have this set of observations, I want to calculate the mean. You can do that manually, but statistics is a module which allows us to calculate means. You simply have to say `import statistics` in Python And `statistics dot mean` will give you the mean of all the numbers you pass in as a list which gives me 6. It means that this these numbers are revolving, or they are basically cantered around 6:00.

Another measure is median. Median is nothing but the middle most number. When you arrange the numbers in a sorted order, be it descending or be it ascending, doesn't matter. You might ask me that. What's the difference between mean and median? Imagine a list of numbers where you have extreme values. In such cases, what you will observe is that the mean would shift towards the extreme values. We call them as an outlier. In such cases when you have extreme values or skewed distributions, we tend to use median which gives me a proper value of the central tendency of the data. If I take the same example and I calculate the median, it would be 6.5.

Similarly, what is mode? Mode is a value which appears most frequently in a data set. Imagine if I ask you to calculate the mean of all the types of clothes. You know the types of clothes is a categorical data and you cannot calculate mean on a categorical data. So, in such cases when you have a categorical data you use mode which is nothing but the most occurring variable. How do you do that? Simply say `statistics mode` and you pass the values. You know 7 is the value, which is occurring twice, so it gives you 7.

Another such measure which talks about the spread of the data is standard deviation. There are many other measures. There is a range, there is a something called as interquartile range, etcetera. Standard deviation is this most appropriate one for all of them. It talks about how spread, how dispersed the data is around your mean. If your if your data has a low standard deviation, it means that the data is not widely spread, it is more concentrated, it is more dense, clustered around the mean, and high standard deviation means it is more spread out. When do you use standard deviation? When you want to study the distributions with, we tend to use standard deviation for that. You will see that in the context of machine learning; there are several places.

For example, there is something called as cross validation. You will see that in the coming sessions which uses standard deviation to validate the performance of your machine learning model. How do we calculate that? It's very simple, straightforward statistics, standard deviation, we pass on the values. It gives you the value 2.41. It says that this has a

spread of 2.41. Another measure is variance. We try to take the average of squared deviations from the mean. It's also a measure to tell the spread in your data set.

And we tend to use variances again in cases where we want to study the variance of the standard deviation, the variance of your data in fact. So, if you see here, investors use variance to see how much risk and investment carries and whether it will be profitable. So how do you calculate the variance? It is simply statistics that variance. You pass on the same set of data, and you will have the result 5.8.

Another measure is correlation. This is called as measure of relatedness. That is how the two variables in your data set are correlated. How do they change with each other? If one grows, what happens to the other? Does it grow or does it decay? And this is one of the bivariate statistics. Whatever we have seen till now was univariate applicable on a single variable. But in correlation you will have two set of values. So, there is a there is something called as correlation coefficient which ranges between -1:00 and 1:00. One means they are strongly correlated.

A classic example is the price of a house verses the area of the house. The more the area increases, the more the price increases. Then you have 0 which says no relationship at all, something which is not correlated at all. The number of marks scored by a student is irrespective of the distance to the school. Irrespective does not matter. 0 correlation -1 Correlation is where you have a negative inversely proportional relationship. For example, the price of a house is inversely proportional to the distance of the house from the city sector. The more the distance the lesser the price. So, this is the kind of bivariate analysis which the correlation coefficient indicates us.

There are certain other measures you will hear about. There is something called as IQR as I talk about variance and standard deviation, IQR is also one of the measures to understand the spread of your data. So, it will be the range of your middle 50% of your sample. So, you separate your data into quartiles, 4 quartiles and the distance between the first quartile and the third quartile is what? What is your IQR? So, in a way IQR skips the extreme values while measuring the range skewness.

We saw skewness is nothing but the direction in which you have a tail in your distribution. So, if you see here, you have a tail towards the left, so it's a left skewed or negatively skewed. Here you have a tail towards the right. So, this is positive skewed or right skewed. Here you don't have the tail. This is a perfect distribution, no skew, and we always expect to have a distribution which has no skew. When do you observe A skew? When you have outliers, when you have outliers towards the left, you will see this left tailed. When you have outliers towards the right, you will see right tailed. And we don't want these tailed observations.

Another measure is kurtosis. Kurtosis is nothing but how heavy your tail is. So, the more the dense these tails are, the larger the kurtosis. So, the distribution weight of a distribution tail relative to the centre of the distribution means how heavy your distribution tail is or how flattened your centre of the distribution is.

And finally, we have outlier. Outlier is nothing but any unusual value. Any abnormal weird value which is lying away from the distribution could be called as outliers.

Thank you.